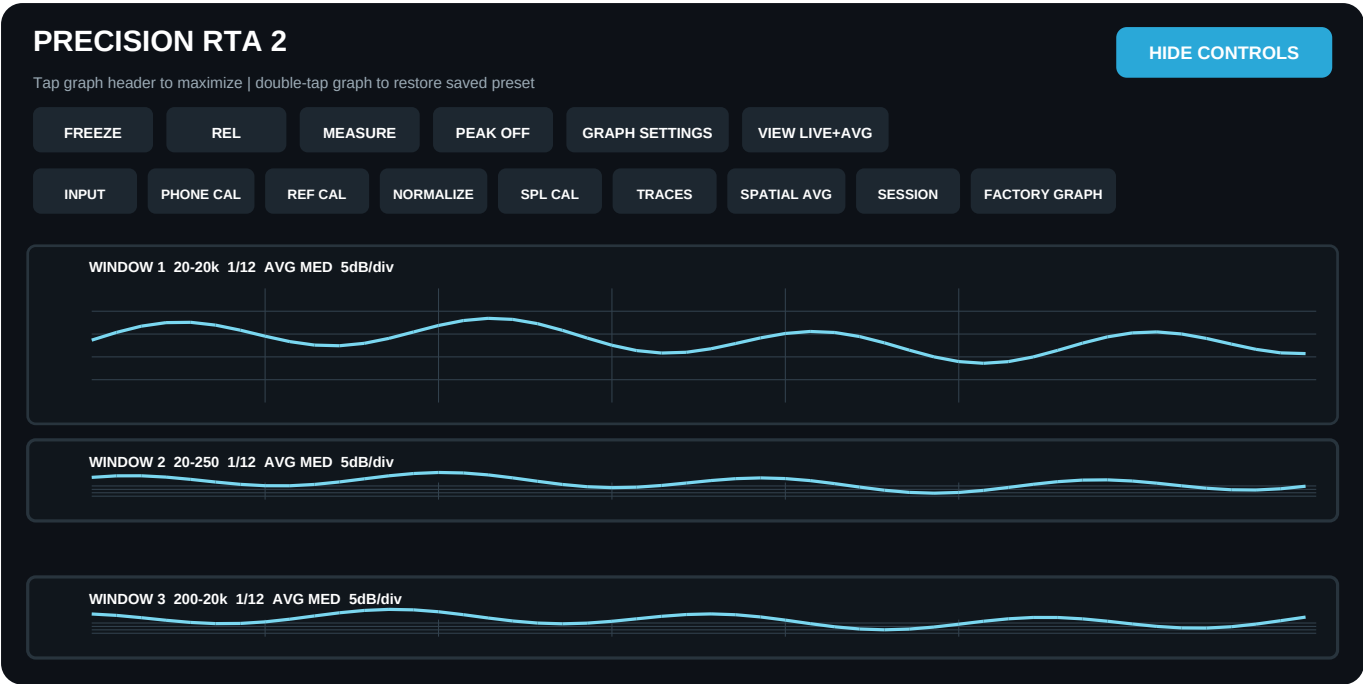


PRECISION RTA

Version 2 User Manual

Measurement-first Android real-time analyzer for car-audio tuning



Purpose of this manual: explain every V2 control, show how the measurement system works, and give a repeatable workflow for collecting measurements that are useful for tuning rather than merely producing attractive graphs.

Target device: Android phone, including the Samsung Galaxy S25 Ultra used to validate the original Precision RTA concept. V2 uses a separate package identity so V1 can remain installed as a fallback during testing.

Manual revision: 2.0 release-candidate documentation. Auto EQ, target-curve editing and phase/crossover optimization are intentionally reserved for a later version after V2 measurement behavior is validated on hardware.

Important: Precision RTA 2 is an acoustic magnitude/RTA tool. It does not currently capture an impulse response or acoustic timing reference. It can reveal response shape, peaks, dips and repeatability, but it cannot directly calculate acoustic phase, excess group delay or exact delay from a magnitude trace alone.

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1. What Precision RTA 2 measures

Precision RTA 2 continuously captures microphone audio, transforms it into the frequency domain, groups energy into fractional-octave bands, and displays the result on logarithmic frequency-response graphs. For formal saved measurements, V2 temporarily raises the base capture to 1/48-octave resolution so the stored data can later be displayed with coarser smoothing without re-recording the vehicle.

The app is optimized around **relative frequency response**. That is usually the useful quantity when tuning a vehicle: how much louder or quieter one part of the spectrum is compared with another. Absolute SPL is available only after a separate one-point level calibration.

1.1 Display processing is non-destructive

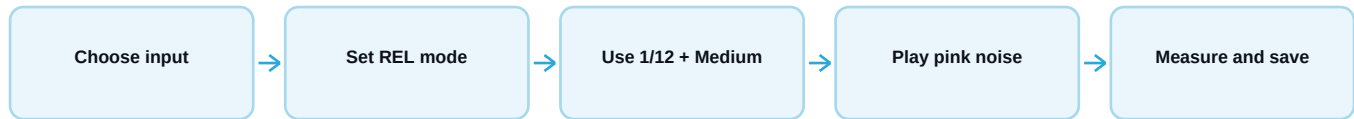
Changing a graph from 1/12 to 1/6 octave, changing the normalization region, hiding a trace, zooming the axes, or changing the dB grid does not rewrite the stored measurement. Those are display operations. This is intentional so a V2 session can later be reused by the V3 EQ optimizer.

Key distinction: smoothing changes how finely the response is displayed; temporal averaging changes how successive measurements are combined over time; normalization shifts the entire trace vertically. None of those should be confused with microphone calibration.

1.2 What it does not measure yet

V2 does not generate a logarithmic sweep, record an impulse response, or use a timing reference. Therefore it should not be used to claim an exact acoustic phase angle, exact propagation delay, impulse response, waterfall or group-delay result. Separate subwoofer / door / combined magnitude measurements can still reveal poor summation, but exact phase/delay recommendations belong to the planned timing-capable workflow.

2. Five-minute quick start



1. Open **INPUT** and verify that the intended microphone is selected. For the built-in phone microphone, Android default is normally appropriate. For an external 3.5 mm measurement microphone connected through USB-C, select the USB/wired input that corresponds to the adapter.
2. Use **REL** mode for vehicle tuning. The default normalization region is 500 Hz to 2 kHz. This makes the graph describe tonal shape instead of arbitrary microphone sensitivity.
3. For general tuning, set the main graph to **1/12 octave** and **Medium averaging**. Use finer 1/24 or 1/48 views to investigate a suspicious feature, not automatically to EQ every narrow ripple.
4. Play steady pink noise through the system at a repeatable level. Avoid clipping the source, head unit, DSP or amplifier. Keep the microphone at ear height/orientation and keep your body out of the direct path as much as practical.
5. Press **MEASURE**. V2 automatically uses a 1/48-octave base capture for the saved measurement even if the graph is displaying 1/12. Let it run long enough to settle; 10-20 seconds is a practical starting point in a car, then press **STOP + KEEP**.
6. Move the microphone a few inches to another nearby listening position and repeat. Keep each individual trace. After several positions, press **SPATIAL AVG** to create an additional equal-energy average.
7. Open **TRACES** to rename measurements and show/hide them. Use **SESSION** to name and save the project once the set is worth keeping.

Do not chase one very narrow dip. If a deep notch moves or disappears when the microphone moves a few inches, it is likely dominated by acoustic interference. Boosting it aggressively can waste headroom without fixing the listening area.

3. Main interface



The graph area always contains three windows. Window 1 receives the larger portion of the vertical space; Windows 2 and 3 provide simultaneous zoomed views. Each window has its own frequency range, vertical scale, smoothing, averaging and grid settings.

3.1 Top-row controls

Control	Function
FREEZE / RESUME	Stops or resumes live graph updates without deleting saved traces.
REL / SPL / dBFS	Cycles display mode. SPL appears only after one-point SPL calibration.
MEASURE	Starts a formal measurement. While measuring it becomes STOP + KEEP.
PEAK OFF / ON	Enables a decaying peak trace. Peaks hold briefly, then decay.
GRAPH SETTINGS	Edits the currently selected graph window.
VIEW LIVE+AVG / VIEW MEASURE ONLY	Chooses whether the live trace remains visible while a formal measurement average builds.

3.2 Second-row controls

Control	Function
INPUT	Select the preferred Android audio input.
PHONE CAL	Enable/import/select/create/verify phone response calibration.
REF CAL	Import or clear the manufacturer calibration for the external reference microphone.
NORMALIZE	Choose the frequency region that equals 0 dB in REL mode.
SPL CAL	Set one global absolute-level offset from an external SPL reading.
TRACES	Show/hide, rename, delete or clear saved traces.
SPATIAL AVG	Create an equal-energy average of the individual measurements in the current session.
SESSION	Save/load session, set project/session names, or discard traces.
FACTORY GRAPH	Restore the selected window to the original V2 defaults and replace its saved preset.

4. Graph interaction and scaling

Pinch

Zoom X and Y. Vertical zoom stays centered on the current dB range.



One-finger drag

Moves the persistent frequency/dB cursor only; it does not rescale the graph.



Tap / double-tap

Tap the header to maximize. Double-tap the graph to restore its saved preset.



4.1 Selecting a window

Touching a graph makes it the selected panel. GRAPH SETTINGS and FACTORY GRAPH act on that selected window. The header text shows the active frequency span, smoothing, averaging and dB-per-division value.

4.2 Pinch zoom

Use two fingers on a graph. A mostly horizontal pinch changes the logarithmic frequency span. A mostly vertical pinch changes the visible dB span. A diagonal pinch can change both. Vertical scaling remains centered around the current dB range, as requested, rather than jumping around the finger location.

4.3 Cursor

A single tap or one-finger drag places a persistent cursor. Dragging is reserved for the cursor and does not alter graph scale. The cursor reports frequency and level from the live trace when available; if live data is unavailable it can fall back to the current measurement preview or a visible saved trace.

4.4 Maximize and restore

Tap within the graph header to maximize that window. Tap the same header again to return to the three-window view. Double-tap inside a graph to restore its saved preset, including frequency limits and vertical limits.

Current state vs saved preset: ordinary edits and pinch zoom are remembered as the current graph state. The explicit Save preset action stores the reset point used by double-tap. Factory Graph replaces both with the original V2 defaults.

5. Graph Settings

Open GRAPH SETTINGS after selecting the desired window. Each window is independent.

5.1 Frequency minimum and maximum

Enter any practical frequency range from about 10 Hz to 24 kHz. The software sanitizes impossible combinations. For car audio, typical investigative ranges include 20-120 Hz for subwoofer work, 50-500 Hz for midbass/crossover investigation, and 20 Hz-20 kHz for overall tonal balance. You do not need to create fixed presets; any range may be entered.

5.2 Y minimum and maximum

The permitted vertical limits extend from -120 dB to +140 dB. In REL mode, a starting range of -30 to +30 dB is usually useful. In calibrated SPL mode, choose a range appropriate for the actual listening level. If a high-volume measurement disappears above the chart, raise Y Max; if detail is visually compressed, reduce the total span.

5.3 Y-axis division

Division	Best use
1 dB	Fine comparison after the scale is already tightly framed.
2 dB	Detailed tuning with less visual crowding.
3 dB	Useful acoustic reference; obvious but not excessively dense.
5 dB	Recommended general-purpose default.
10 dB	Large dynamic ranges or quick overview.

Grid division and Y Min/Y Max are independent. A 1 dB grid does not force a narrow total span, and a 10 dB grid does not change the measurement itself.

5.4 Minor grid and labels

Minor grid lines can be enabled for additional visual reference. Y-axis labels can be disabled when a small window becomes cluttered. The intent is to preserve readable traces rather than maximize the number of numbers on screen.

6. Fractional-octave smoothing

Each graph can independently display 1/3, 1/6, 1/12, 1/24 or 1/48-octave data. Finer settings reveal increasingly narrow structure. Coarser settings make broad tonal trends easier to interpret.

Setting	Interpretation	Recommended use
1/3	Very broad tonal view	Quick balance checks; intentionally hides narrow structure.
1/6	Broad but useful	General EQ trend and sanity checking.
1/12	Detailed without being extreme	Recommended V2 tuning default.
1/24	Fine diagnostic view	Investigate resonances, crossover structure and narrow peaks.
1/48	Very fine diagnostic view	Confirm exact shape/location of features; saved measurement base resolution.

Important: 1/48 is not automatically "more correct" for EQ. It is more detailed. A car cabin contains comb filtering and position-dependent notches that become extremely obvious at fine resolution. The planned V3 optimizer will use a deliberately chosen analysis smoothing rather than blindly flattening every 1/48-octave variation.

REW likewise exposes smoothing from 1/3 through 1/48 octave and recommends variable/coarser treatment for equalization rather than assuming the finest visible structure should always be corrected. See the references at the end of this manual.

7. Temporal averaging and peak hold

7.1 Averaging choices

Mode	Approx. time constant	Use
Fast	0.15 s	Immediate feedback while making an adjustment.
Medium	0.5 s	Recommended live tuning default.
Slow	1.5 s	More stable display in noisy conditions.
Very slow	4 s	Steady overview when response changes slowly.

Live graph averaging is performed in the energy/power domain before conversion back to dB. This avoids the bias that would result from simply averaging logarithmic dB numbers.

7.2 Formal measurement averaging

Formal MEASURE mode also accumulates response energy rather than arithmetic dB values. The measurement accumulator is independent from the cosmetic temporal averaging setting of the graph. This makes saved measurements better suited to later EQ analysis.

7.3 Peak hold

PEAK ON draws a peak trace. A peak is retained briefly; if it is not exceeded, it begins to decay rather than remaining forever. Peak hold is useful for finding intermittent resonances or confirming the highest excursion of a changing response. It is not used as the formal averaged measurement.

8. REL, dBFS and SPL modes

8.1 REL - recommended for tuning

REL normalizes the current response to an energy-average level within a chosen reference frequency range. The default region is 500 Hz to 2 kHz. The result describes frequency-response shape around a 0 dB reference and avoids implying that an uncalibrated phone knows absolute sound pressure.

8.2 dBFS-like display

dBFS-like mode shows the analyzer level relative to the digital capture chain. It is useful for checking whether input level changes, but it is not sound pressure level.

8.3 SPL

SPL mode becomes meaningful only after SPL CAL is performed. V2 applies one global level offset; frequency-response shape correction remains the job of PHONE CAL. This separation prevents an absolute-level adjustment from being confused with tonal calibration.

Recommended rule: tune frequency response in REL mode. Use SPL mode when you specifically need a level reference and have calibrated it against a known meter or calibrator.

9. Normalization

NORMALIZE defines the frequency interval whose energy-average level becomes 0 dB in REL mode. Stored measurement values are not rewritten. You can reopen the same session and choose a different normalization range later.

9.1 Why normalize over a range instead of one frequency

Using a range makes the reference less sensitive to one narrow peak or dip. The 500 Hz-2 kHz default is broad enough to be stable for many full-range vehicle measurements, but it is not mandatory.

9.2 When to change it

For a subwoofer-only measurement, the default midrange reference is inappropriate because there may be no meaningful signal there. Choose a reference inside the measured driver's useful band, for example a stable portion of 40-80 Hz. For midbass-only work, choose a stable region inside the midbass band.

Changing normalization only moves the whole trace vertically. It cannot create, remove or repair a peak/dip. If the shape changes after normalization, something else changed.

10. Formal Measurement mode

The live RTA is useful for watching changes. Formal MEASURE mode is used when you want a trace worth saving, comparing or averaging.

10.1 Start and stop

Press MEASURE to begin. V2 forces the analyzer base to 1/48-octave resolution for the capture so the saved trace retains fine frequency information even if all three windows are displaying 1/12 or 1/6. Press STOP + KEEP when the capture is complete.

10.2 No artificial minimum duration

The app does not prevent you from stopping early. However, 1/48 base capture uses a long FFT window (262,144 samples, approximately 5.46 seconds at 48 kHz), so extremely short captures are not useful. In a vehicle, 10-20 seconds of steady pink noise is a practical baseline when conditions are stable.

10.3 Live + Average versus Measurement Only

VIEW LIVE+AVG keeps the live response visible while the averaged measurement builds. **VIEW MEASURE ONLY** hides the live response during capture so you can concentrate on the stable accumulated result. The selected behavior is remembered.

10.4 Transient rejection

V2 conservatively rejects frames whose broadband level deviates strongly from the established capture level. The status line reports approximately how much time was kept and discarded. The detector is not trying to identify speech, a door slam or a horn by name; it simply protects the average from obvious broadband disturbances.

If a real system adjustment is made during the same measurement, the transient detector may not understand your intent. For before/after tuning, stop the measurement, make the adjustment, then start a new measurement.

11. Traces, spatial averages and sessions

11.1 Trace manager

TRACES opens management options for show/hide, rename, delete or clear. Each individual measurement remains independent. Hiding a trace never deletes its data.

11.2 Spatial average

SPATIAL AVG uses the individual non-spatial-average traces in the current session and creates an additional equal-energy average. The original traces remain available and can still be toggled independently.

11.3 Why spatial averaging matters in a car

The response at one exact point near your ear can contain very narrow interference notches that change dramatically a few centimeters away. A small cluster of measurements around the listening position helps separate persistent tonal problems from position-specific comb filtering.

11.4 Sessions

SESSION saves the project name, session name, all traces, graph configurations, normalization range and display mode. Unsaved active work is also kept in a temporary recovery file after measurements so an Android process kill does not automatically destroy the session. A recovery prompt appears when appropriate.

Suggested naming: Project = vehicle/system. Session = state of the tune. Trace = measurement position or condition.
Example: Project "Versa" / Session "DSP pass 2" / Traces "Driver center", "Driver left +2 in", "Driver right +2 in".

12. Selecting the microphone input

INPUT asks Android for the available recording devices and lets you set a preferred input. The status line reports the routed input when Android exposes it.

12.1 Built-in phone microphone

Android default is the safest starting point for the S25 Ultra unless you are deliberately selecting a particular exposed built-in device. Precision RTA requests Android's UNPROCESSED audio source when the device reports support; Android documents UNPROCESSED as a microphone source intended for unprocessed/raw sound when available, with fallback behavior otherwise.

12.2 External 3.5 mm microphone through USB-C

A USB-C audio adapter must actually expose a microphone input to Android. Some dongles are output-only; others expect a TRRS headset microphone and may not provide the bias/preamp needed by every measurement microphone. If the adapter is recognized, INPUT should show a USB or wired input. Confirm the status line after selection.

Routing warning: Android ultimately controls device routing. The app can request a preferred device, but it cannot guarantee that every Samsung/USB audio combination exposes every physical microphone path. Always verify the routed-input label before trusting a calibration profile.

13. Phone microphone calibration

PHONE CAL corrects frequency-response shape. It is separate from SPL CAL, which only changes absolute level.

13.1 Importing a phone calibration file

V2 accepts a text calibration file with at least two numeric columns: frequency and dB value. Header/comment lines are ignored. The import dialog asks whether the second column represents measured microphone error to subtract or correction values to add. Additional columns, including phase, are ignored in V2.

13.2 Multiple profiles

Profiles can store device model, input name, sample rate and orientation metadata. This matters because a phone microphone and its acoustic openings are not perfectly omnidirectional. A profile made in portrait orientation should not silently be assumed identical to landscape. V2 warns when the active profile orientation and current orientation differ and can point out another saved profile that matches.

13.3 Enable/disable

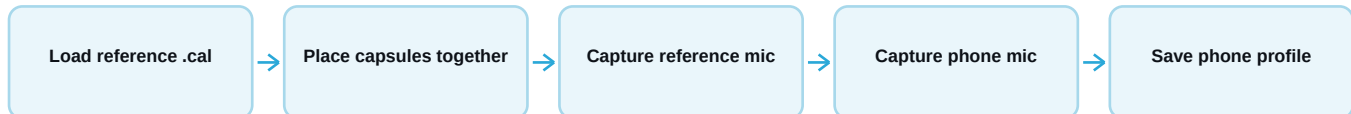
A saved profile can remain selected while correction is temporarily disabled. This is useful for A/B verification without deleting the profile.

14. Reference microphone calibration and Auto Cal

14.1 REF CAL

REF CAL stores the manufacturer response-correction file for the external reference microphone. If the microphone manufacturer supplies orientation-specific files, use the file intended for the way the capsule is pointed. For example, miniDSP publishes separate on-axis and 90-degree calibration files for its UMIK microphones; the same general principle applies to any microphone whose supplier documents orientation-specific calibration.

14.2 Reference-match Auto Cal workflow



1. Load the manufacturer reference-microphone calibration through REF CAL if one is available.
2. Put the reference capsule immediately beside the phone microphone opening and point both in the same direction. Keep the phone, speaker and microphones fixed.
3. Play steady pink noise from one speaker. Use enough level for strong signal-to-noise ratio without clipping.
4. PHONE CAL -> Reference-match Auto Cal -> capture the external reference microphone.
5. Without changing source level or geometry, capture the built-in phone microphone. The app derives the difference in response shape and removes the overall midband sensitivity offset.
6. Save/use the profile, then run Verify active calibration with a fresh reference/phone pair.

Sequential calibration is sensitive to anything that changes between captures. A moved phone, moved source, changed volume, HVAC noise or a body moving near the microphones can become part of the correction curve. Treat geometry control as seriously as the software steps.

15. Calibration verification

Verify active calibration performs a new external-reference capture followed by a new phone capture. V2 calculates the RMS mismatch from 50 Hz to 10 kHz after removing the overall level/sensitivity difference. This is intentional: Auto Cal is a frequency-response-shape calibration, not an absolute SPL calibration.

15.1 Interpreting the score

Fresh-match RMS error	Interpretation
Below about 1 dB	Excellent practical agreement if the setup was genuinely unchanged.
About 1-2 dB	Generally useful for vehicle frequency-response work; inspect the overlay/shape if available.
Above about 2 dB	Repeat the procedure and investigate position/orientation/routing/noise before trusting the profile.

Those bands are practical guidance, not a laboratory uncertainty specification. The score includes errors from the sequential capture process and the acoustic environment.

15.2 Common causes of a poor verification

Wrong Android input, different reference calibration convention, moving the microphones, using an orientation-mismatched calibration file, changing playback volume, insufficient signal-to-noise ratio, USB adapter behavior, or environmental noise.

16. SPL calibration

SPL CAL applies a single global offset so the analyzer can display an approximate absolute SPL reference. It does not create a frequency-response correction curve.

1. Use a known external SPL meter/reference and put it at the same acoustic position as the phone.
2. Play a steady 1 kHz tone at a safe, stable level.
3. Wait for the Precision RTA live spectrum to settle.
4. Open SPL CAL and enter the external reference reading.
5. Cycle the mode button to SPL and confirm that repeated measurements are plausible.

The one-point SPL feature is useful for practical level reference, but Precision RTA 2 is not a standards-certified sound-level meter. Phone microphone sensitivity, adapter gain and Android routing can change outside the conditions under which you calibrated it.

17. Accurate car-audio measurement workflow

This chapter is the recommended procedure when the goal is to make EQ decisions from Precision RTA rather than simply inspect a live waveform.

17.1 Prepare the vehicle and signal chain

1. Park in a repeatable environment. Close doors/windows. Decide whether the engine/HVAC should be off for measurement purity or on to represent real driving noise; for EQ response work, off is usually the cleaner baseline.
2. Document the system state: head-unit volume, loudness/bass boost, DSP preset, crossover settings, time alignment, amplifier gain, EQ and source. If you cannot disable an effect, at least keep it unchanged between measurements.
3. Avoid source or amplifier clipping. More level is not automatically more accurate. The goal is strong signal-to-noise ratio with clean headroom.
4. Select the intended microphone input and calibration profile before starting the measurement set.
5. Use the same phone/microphone orientation for every trace that will be compared.

17.2 Microphone location

For a driver-seat tune, start near the midpoint between your ears at normal ear height. Do not hold the microphone directly in your hand if that causes it to move or lets your hand/body block the capsule. A small fixture or headrest mount improves repeatability.

After the center measurement, use a small cluster around the listening position. A practical five-position set is center, a few inches left, a few inches right, a few inches forward and a few inches rearward. The exact distance is less important than keeping the cluster representative of where your head actually moves.

18. Capturing a tuning set

18.1 Full-system tonal balance

1. Play steady pink noise through the full system.
2. Use REL mode, 1/12-octave display smoothing and Medium averaging as the starting view.
3. Set the main window to a useful visible range, commonly -20 to +20 or -30 to +30 dB in REL mode. Choose 3 or 5 dB divisions.
4. Press MEASURE. Leave the microphone fixed and let the capture run roughly 10-20 seconds under stable conditions.
5. Press STOP + KEEP and rename the trace for that position.
6. Repeat at the remaining positions without changing playback level or system tuning.
7. Create SPATIAL AVG and compare it with the individual traces.

18.2 Decide what is real before EQ

A broad peak that appears in every nearby position is a strong EQ candidate. A very narrow dip that changes frequency or depth dramatically between positions is a poor candidate for boost. A broad dip that persists in all positions may be a real balance/crossover problem, but still consider speaker/crossover/phase causes before applying large boost.

Recommended hierarchy: fix hardware/crossover/polarity/placement problems first; use EQ for repeatable tonal excess/deficiency; avoid spending headroom trying to fill a position-specific acoustic null.

19. Driver and crossover investigation

Although V2 does not yet calculate phase or delay, separate magnitude traces are extremely useful preparation for crossover work.

19.1 Recommended capture set

Trace	System state	Purpose
Sub only	Door speakers muted	Establish subwoofer magnitude response.
Doors only	Subwoofer muted	Establish front-system/midbass response.
Sub + doors	Normal crossover state	Observe acoustic summation through crossover.
Polarity test (optional)	One section polarity reversed	Check whether summation improves or worsens around crossover.

19.2 Interpreting summation

If sub-only and door-only traces each have useful output around the crossover but the combined trace becomes substantially lower in the overlap region, destructive interference is likely. V2 can reveal that magnitude symptom. It cannot uniquely calculate the exact phase-dial angle from that symptom alone.

A conservative troubleshooting sequence is: verify polarity, try available delay/time-alignment changes, try a nearby crossover frequency/slope, and remeasure after each deliberate change. Save each condition as a separate trace/session rather than modifying several variables at once.

20. Troubleshooting

Symptom	Likely cause / action
Trace is above or below the visible chart	Pinch vertically or enter new Y Min/Y Max in Graph Settings. The measurement is not lost just because it is off-screen.
Graph looks too flat / too dramatic	Check Y-axis span and dB/division. Visual slope can be misleading when scale changes.
1/48 response feels slow	Expected. Formal 1/48 capture uses a 262,144-sample FFT (~5.46 s at 48 kHz). Use 1/12 or 1/6 for faster live feedback.
External mic does not appear	Confirm the USB-C adapter supports microphone input and the mic/adapter wiring is compatible. Reopen INPUT and verify Android exposes it.
Status shows the wrong input	Do not trust the calibration. Reselect input/reconnect adapter and confirm routed-input text.
Calibration verification score is poor	Repeat with fixed geometry, correct manufacturer cal convention, correct orientation, higher clean signal-to-noise ratio and stable playback level.
One deep notch moves when mic moves	Treat it as likely position-dependent cancellation; do not automatically boost it.
Measurement discarded a lot of time	Environment or playback level was unstable. Re-run with fewer sudden disturbances.
App closes / Android kills it	The current trace set may be offered for temporary recovery on next launch.
Double-tap changes graph unexpectedly	It restores the saved preset, not necessarily the last pinch state. Use Save preset when the desired reset state is configured.

21. Recommended defaults and quick reference

Task	Suggested starting settings
General vehicle tonal tuning	REL; 1/12 octave; Medium averaging; 3 or 5 dB/div; 20 Hz-20 kHz.
Subwoofer response	REL; 1/12 or 1/24; Medium/Slow; 10-200 Hz; normalization inside sub band.
Investigate a narrow resonance	1/24 or 1/48; tight frequency window; 1-3 dB/div if the Y span permits.
Live EQ adjustment	1/6 or 1/12; Fast or Medium averaging.
Formal measurement for saving	MEASURE; app stores 1/48 base automatically; 10-20 s practical starting duration.
Compare multiple seat/head positions	Keep individual traces; equal level/settings; make Spatial Average after capture.
Check absolute level	Perform SPL CAL first; use SPL only as a practical calibrated reference.
Phone calibration	Use manufacturer-calibrated reference mic when possible; keep geometry/orientation fixed; Verify afterward.

21.1 Default graph windows

Window	Frequency	Y range	Smoothing	Average	Grid
1	20 Hz-20 kHz	-30 to +30 dB	1/12	Medium	5 dB
2	20-250 Hz	-30 to +30 dB	1/12	Medium	5 dB
3	200 Hz-20 kHz	-30 to +30 dB	1/12	Medium	5 dB

22. Data integrity and session behavior

A saved MeasurementTrace contains frequency/level arrays plus measurement metadata such as capture time, accepted/rejected frames and time, input label, sample rate, calibration profile and orientation. Sessions also retain graph configuration, normalization and display mode.

22.1 Why formal measurements use 1/48 base capture

A coarse saved trace cannot later be made genuinely finer by interpolation. V2 therefore increases the acquisition base to 1/48 during MEASURE even when the visible graph is coarser. Later display operations can combine fine bands into 1/24, 1/12, 1/6 or 1/3 without pretending detail exists that was never recorded.

22.2 What remains non-destructive

Show/hide, cursor, graph zoom, Y scale, dB grid, normalization and display smoothing do not delete or rewrite the underlying trace. Rename changes metadata only. Delete/clear are the operations that remove a trace from the active session.

22.3 Session recovery versus explicit save

Temporary recovery protects active work from accidental process loss; it is not a substitute for an explicit SESSION save. Save a session when you want it to become a deliberate tuning checkpoint.

23. Limitations and V3 boundary

Capability	V2 status
High-detail RTA magnitude response	Implemented
Independent three-window display	Implemented
Saved high-resolution measurements	Implemented
Spatial averaging	Implemented
Phone/reference calibration workflow	Implemented
Absolute level offset	Implemented
Target / house curve editor	Deferred to V3
Automatic PEQ / fixed-band optimizer	Deferred to V3
Poweramp filter export	Deferred to V3
Suspected cancellation review for Auto EQ	Deferred to V3
Predicted corrected response	Deferred to V3
Exact phase/delay measurement	Requires timing/impulse-response capability; not inferred from RTA magnitude alone

Deferring Auto EQ is deliberate. The optimizer will only be trustworthy if the measurement acquisition, averaging, calibration and saved-trace behavior are trustworthy first. V2 is designed to establish that foundation on the actual phone and vehicle.

24. Technical references and further reading

The following external documentation informed the measurement guidance and terminology in this manual. Precision RTA 2 is an independent project and is not affiliated with REW, Android, Samsung or miniDSP.

Room EQ Wizard (REW)

RTA Window: REW describes pink noise as producing a flat RTA response because its energy falls 3 dB per octave while logarithmic RTA bins widen with frequency. It also documents the need for averaging with pink-noise measurements.

https://www.roomeqwizard.com/help/help_en-GB/html/spectrum.html

Graph smoothing: REW exposes 1/3 through 1/48-octave smoothing and notes that extremely fine structure is not automatically the correct basis for equalization.

https://www.roomeqwizard.com/help/help_en-GB/html/graph.html

SPL calibration: REW documents calibrating SPL readings against an external meter/reference.

https://www.roomeqwizard.com/rewv5_help.pdf

Android Developers

MediaRecorder.AudioSource.UNPROCESSED: Android documents this source as tuned for unprocessed/raw microphone sound when available, with fallback behavior otherwise. Precision RTA requests this path when supported.

<https://developer.android.com/reference/android/media/MediaRecorder.AudioSource#UNPROCESSED>

miniDSP measurement-microphone documentation

miniDSP documents separate on-axis and 90-degree calibration files for UMIK microphones and explains that the calibration file must match microphone orientation. This manual uses that as an example of the broader rule: follow the calibration/orientation instructions supplied with your own reference microphone.

<https://support.minidsp.com/support/solutions/articles/47000681633-which-direction-should-i-point-the-umik-1->

25. Field checklist

Before measuring

- Correct input shown in status
- Correct phone/reference calibration selected
- Consistent microphone orientation
- Stable source/head-unit/DSP volume
- No clipping or obvious AGC behavior
- REL mode for tonal tuning
- 1/12 + Medium as the general starting view

For each saved trace

- Same system settings unless the trace deliberately tests a change
- Microphone fixed during the capture
- Roughly 10-20 s stable pink-noise capture as a practical starting point
- Review discarded-time counter
- Rename the trace immediately so its position/condition is unambiguous

Before applying EQ

- Compare several nearby positions
- Create a spatial average
- Identify broad, repeatable problems first
- Treat narrow moving nulls as acoustic interference until proven otherwise
- Fix polarity/crossover/placement problems before spending large EQ boosts
- Save a baseline session before changing the tune

End of Precision RTA 2 User Manual